

CE 206: Engineering Computation Sessional

Eigenvalues and Eigenvectors

Eigenvalue and Eigenvector

Definition. Let A be an $n \times n$ matrix. A number $\lambda \in \mathbb{R}$ is called an **eigenvalue** of the matrix A if $A\mathbf{v} = \lambda\mathbf{v}$ for a nonzero column vector $\mathbf{v} \in \mathbb{R}^n$.

The vector $\mathbf{v}$ is called an **eigenvector** of A belonging to (or associated with) the eigenvalue λ .

Remarks. • Alternative notation:
eigenvalue = **characteristic value**,
eigenvector = **characteristic vector**.

• The zero vector is never considered an eigenvector.

Mathematics of eigenvalues

Homogeneous systems: $[A] \{x\} = 0$

Trivial Solution: $\{x\} = 0$

Non-trivial solutions are possible, but generally they will not be unique.

Eigenvalue problems involve a special set of these nontrivial solutions, which has the form

$$[[A] - \lambda[I]] \{x\} = 0$$

For this case, there could be a value of λ that makes the equations equal zero. This is called an *eigenvalue*.

How to determine eigenvalues?

The homogeneous system of linear equations:

$$(a_{11} - \lambda) x_1 + a_{12} x_2 + a_{13} x_3 = 0$$

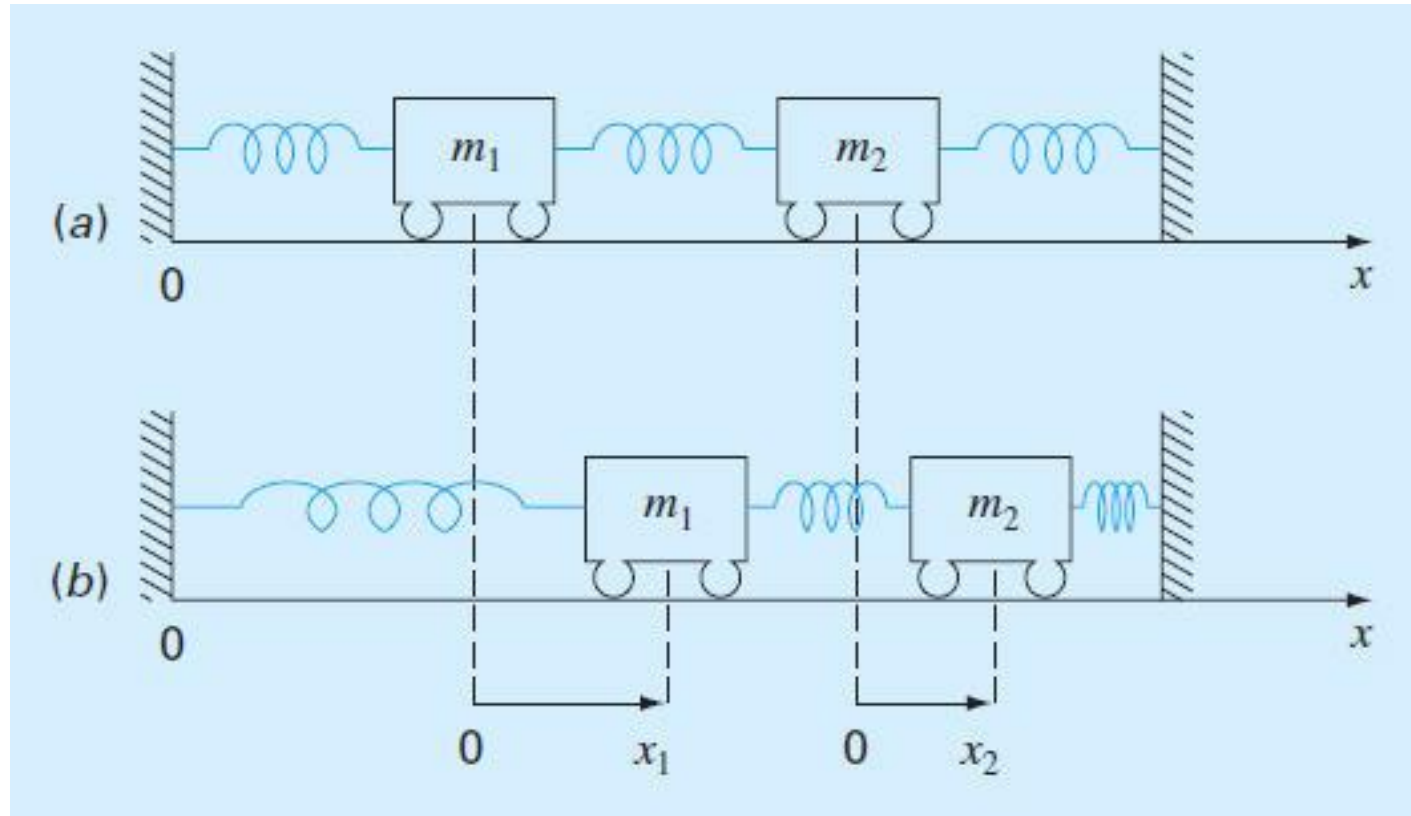
$$a_{21} x_1 + (a_{22} - \lambda) x_2 + a_{23} x_3 = 0$$

$$a_{31} x_1 + a_{32} x_2 + (a_{33} - \lambda) x_3 = 0$$

$$[[A] - \lambda[I]] \{x\} = 0$$

Set $\det [[A] - \lambda[I]] = 0$, Solve for λ from the characteristic polynomial.

Oscillations in a mass-spring system



$$m_1 \frac{d^2 x_1}{dt^2} = -k x_1 + k(x_2 - x_1)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1) - kx_2$$

Solving strategy

Assume a sinusoidal solution $x_i = X_i \sin(\omega t)$ where $\omega = \frac{2\pi}{T_p}$

$$\left(\frac{2k}{m_1} - \omega^2\right) X_1 - \frac{k}{m_1} X_2 = 0$$

$$\frac{k}{m_2} X_1 + \left(\frac{2k}{m_2} - \omega^2\right) X_2 = 0$$

Given: $m_1 = m_2 = 40 \text{ kg}$; $k = 200 \text{ N/m}$

$$(10 - \omega^2) X_1 - 5 X_2 = 0$$

$$-5 X_1 + (10 - \omega^2) X_2 = 0$$

The eigenvalue represents the square of the fundamental frequency.

Solution strategy

Evaluate the determinant to yield a polynomial

$$\begin{vmatrix} 10 - \omega^2 & -5 \\ -5 & 10 - \omega^2 \end{vmatrix} = (\omega^2)^2 - 20\omega^2 + 75$$

The two roots of this "characteristic polynomial" are the system's eigenvalues:

$$\omega^2 = 5 \text{ /s}^2$$

$$\omega = 2.236 \text{ /s}$$

$$T_p = 2\pi/2.236 = 2.81 \text{ s}$$

$$X_1 = X_2$$

$$V = \begin{bmatrix} -0.7071 \\ -0.7071 \end{bmatrix}$$

$$\omega^2 = 15 \text{ /s}^2$$

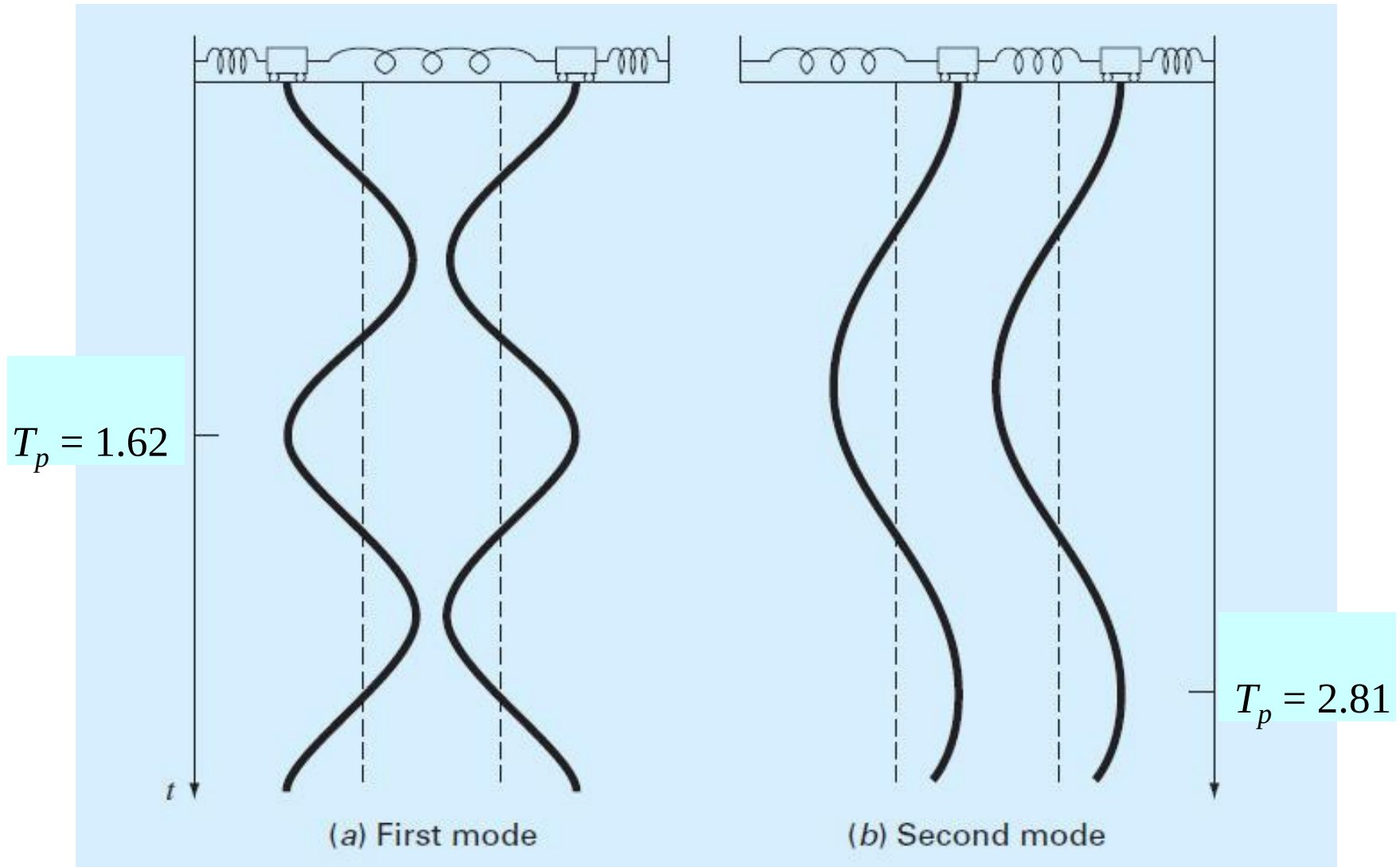
$$\omega = 3.873 \text{ /s}$$

$$T_p = 2\pi/3.373 = 1.62 \text{ s}$$

$$X_1 = -X_2$$

$$V = \begin{bmatrix} -0.7071 \\ 0.7071 \end{bmatrix}$$

Principal modes of vibration



MATLAB built-in function `eig()`

```
>> A = [10 -5;-5 10]
```

```
A =
```

```
    10    -5  
    -5    10
```

```
>> [v,lambda] = eig(A)
```

```
v =
```

```
 -0.7071   -0.7071  
 -0.7071    0.7071
```

```
lambda =
```

```
     5     0  
     0    15
```

Assignment

A mass-spring system represents a model for a three-story building. Use MATLAB to determine the eigenvalues and eigenvectors for this system. Graphically represent the modes of vibration for the structure.

